

Functional Annotation Report: PaaH (PP_3282, UniProt Q88HS1)

3-Hydroxyadipyl-CoA dehydrogenase of *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

PaaH (gene *paaH*; ordered locus **PP_3282**; UniProt **Q88HS1**) of *Pseudomonas putida* KT2440 is a **soluble, cytoplasmic, NAD⁺-dependent 3-hydroxyacyl-CoA dehydrogenase (EC 1.1.1.35)** that acts in the **lower, β -oxidation-like segment of the aerobic phenylacetate (phenylacetyl-CoA) degradation pathway**. Its primary catalytic function is the stereospecific oxidation of **(S)-3-hydroxyadipyl-CoA to 3-oxoadipyl-CoA (β -ketoadipyl-CoA)**, coupled to reduction of NAD⁺ to NADH. This is the penultimate redox reaction of the pathway; the resulting 3-oxoadipyl-CoA is thiolytically cleaved to **succinyl-CoA + acetyl-CoA**, which enter the TCA cycle. The identification rests on database/EC annotation, membership in the 3-hydroxyacyl-CoA dehydrogenase (PaaH) family with a confirmed N-terminal Rossmann NAD-binding fold, and functional characterization of the surrounding *paa* pathway in *P. putida* and *E. coli*.

Gene-identity verification: The gene symbol *paaH*, the organism (*P. putida* KT2440), the annotated function (3-hydroxyadipyl-CoA dehydrogenase, EC 1.1.1.35), and the InterPro family/domains (3-OHacyl-CoA_DH_PaaH, IPR011967; NAD-binding and C-terminal domains) are **mutually consistent** with the literature on the bacterial phenylacetate catabolon. No ambiguity or misassignment was detected.

2. Biochemical Function

2.1 Reaction catalyzed



EC 1.1.1.35 is the general "3-hydroxyacyl-CoA dehydrogenase" that oxidizes the C3 hydroxyl of an (S)/L-3-hydroxyacyl-CoA thioester to the corresponding 3-oxo (β -keto) acyl-CoA, generating NADH. The reaction is the third of the four classic β -oxidation reactions (hydration $\rightarrow$ **dehydrogenation** $\rightarrow$ thiolysis) and is freely reversible in vitro [Barycki et al., 2000,].

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In PaaH the acyl substrate is the **C6 dicarboxylic thioester 3-hydroxyadipyl-CoA**, a ring-cleavage product of phenylacetyl-CoA catabolism rather than a straight-chain fatty acyl-CoA.

2.2 Substrate specificity

The substrate is defined by the pathway context and by the enzyme's subfamily assignment: - The InterPro **PaaH subfamily signature (IPR011967)** distinguishes phenylacetate-pathway 3-hydroxyacyl-CoA dehydrogenases from general fatty-acid β -oxidation HADs, indicating specificity for the short-chain **dicarboxylic** 3-hydroxyacyl-CoA intermediate (3-hydroxyadipyl-CoA) generated after aromatic-ring opening [Ferrández et al., 1998,];

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Olivera et al., 1998,].

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- Downstream chemistry constrains the substrate: because the terminal products of the pathway are **succinyl-CoA** + **acetyl-CoA** [Nogales et al., 2007,],

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the immediate PaaH product must be **3-oxoadipyl-CoA (β -keto adipyl-CoA)**, and therefore its substrate is **3-hydroxyadipyl-CoA**. - Family members are stereospecific for the (S)/L-3-hydroxy configuration [Barycki et al., 2000,].

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2.3 Cofactor

NAD⁺ is the electron acceptor. The N-terminal Rossmann dinucleotide-binding domain carries the cofactor; the homologous PaaH-type enzyme RePaaH1 shows "the cofactor-binding site is located on the Rossmann fold" [Kim et al., 2014,].

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3. Structural / Bioinformatic Evidence

Direct sequence analysis of Q88HS1 (this study) confirms the annotation:

- **Length:** 505 residues — consistent with the PaaH subfamily (e.g., *E. coli* PaaH $\approx$ 475 aa), which is larger than the ~300-aa mammalian mitochondrial HAD owing to extra subfamily-specific segments (InterPro IPR041040, 3HCDH_RFF).
- **N-terminal Rossmann NAD-binding fingerprint:** residues 12–22 read VAVIGAGAMGAG, containing the canonical **G-x-G-x-x-G glycine-rich motif (GAGAMG, position 14)** that binds the ADP moiety of NAD(H). This is diagnostic of the dinucleotide-binding fold.
- **Two-domain architecture (by homology):** N-terminal Rossmann NAD-binding domain + C-terminal α -helical dimerization domain, as crystallographically established for the closely related PaaH-family enzyme RePaaH1 [Kim et al., 2014,

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and for L-3-hydroxyacyl-CoA dehydrogenase [Barycki et al., 2000,

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- **Catalytic machinery (by family conservation):** a **His–Glu catalytic dyad** in the interdomain cleft abstracts a proton from the C3 hydroxyl while NAD⁺ accepts the hydride [Barycki et al., 2000,

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Mugo et al., 2015,

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- **InterPro domains (from the target record):** 3-OHacyl-CoA_DH_CS (IPR006180), 3-OHacyl-CoA_DH_NAD-bd (IPR006176), 3-OHacyl-CoA_DH_PaaH (IPR011967), 3HC_DH_C (IPR006108),

3HCDH_RFF (IPR041040) — all congruent with an NAD-dependent 3-hydroxyacyl-CoA dehydrogenase of the phenylacetate pathway.

This constitutes convergent **bioinformatic/structural inference** (sequence motif + family + homologous crystal structures) supporting the annotated function.

4. Pathway Context

PaaH is one enzyme of the **phenylacetyl-CoA catabolon**, an aerobic "hybrid" pathway that funnels many aromatic compounds into central metabolism [Ferrández et al., 1998,

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Luu et al., 2013,

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The pathway, defined genetically in *P. putida* U/KT2440 and *E. coli*, comprises [Olivera et al., 1998,

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Ferrández et al., 1998,

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1. **Activation** — phenylacetate → phenylacetyl-CoA (PaaK, phenylacetate–CoA ligase).
2. **Ring 1,2-epoxidation/hydroxylation** — a multicomponent oxygenase (PaaABC(D)E) acting on the CoA-activated ring.
3. **Ring opening** — a hydratase/aldehyde-dehydrogenase (PaaZ) that cleaves the ring to a linear dicarboxylic-CoA (a 3-oxo-unsaturated adipyl/suberyl-CoA species).
4. **β-oxidation-like processing** — enoyl-CoA hydratase/isomerase, then **PaaH (3-hydroxyacyl-CoA dehydrogenase)**, then a β-ketoadipyl-CoA thiolase.
5. **Termination** — thiolytic cleavage of 3-oxoadipyl-CoA to **succinyl-CoA + acetyl-CoA** [Nogales et al., 2007,

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PaaH's specific position: it performs step 4's dehydrogenation, converting **3-hydroxyadipyl-CoA → 3-oxoadipyl-CoA (β-ketoadipyl-CoA)**. The product is the direct substrate of the terminal thiolase (in *P. putida* shown to yield succinyl-CoA + acetyl-CoA), so PaaH sits immediately upstream of the pathway's convergence with central carbon metabolism and with the β-ketoadipate node. This pathway allows *P. putida* to use phenylacetate as sole carbon and energy source (and it is chemotactically attractive) [Luu et al., 2013,].

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The catabolon integrates upper routes for phenylethylamine/phenylethanol and styrene, all of which converge on phenylacetyl-CoA [Olivera et al., 1998, ;

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Arias et al., 2008, ;

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Alonso et al., 2003,].

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Note on nomenclature: In the *P. putida* KT2440 gene naming (used for PP_3282), *paaH* denotes the 3-hydroxyacyl-CoA dehydrogenase. Homologous *paa* clusters in other organisms use different letters for the same enzymatic steps (e.g., in some *P. putida* studies the terminal thiolase is designated by a different letter), so care is needed when mapping gene letters across strains. The **enzymatic identity of PP_3282 as the 3-hydroxyadipyl-CoA dehydrogenase (EC 1.1.1.35)** is unambiguous.

5. Subcellular Localization

PaaH is a **soluble cytoplasmic (cytosolic) enzyme**. Evidence: - Its substrate and product are CoA thioesters, which are cytosolic metabolites of the intracellular β-oxidation-like module [Olivera et al., 1998,].

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- The 505-aa sequence has **no N-terminal signal peptide and no transmembrane segment**, and the 3-hydroxyacyl-CoA dehydrogenase family is uniformly soluble/globular [Barycki et al., 2000,

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Kim et al., 2014,

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- The enzyme is predicted to function as a **homodimer** (via the C-terminal domain), like characterized family members.

6. Supported and Refuted Hypotheses

Hypothesis	Verdict	Basis
PaaH is an NAD ⁺ -dependent 3-hydroxyacyl-CoA dehydrogenase (EC 1.1.1.35)	Supported	EC/DB annotation + Rossmann motif + family membership
Its physiological substrate is 3-hydroxyadipyl-CoA (→ 3-oxoadipyl-CoA)	Supported (inference)	Pathway position; products are succinyl-CoA + acetyl-CoA [PMID 17259607]
PaaH acts in the β-oxidation-like lower segment of the phenylacetate catabolon	Supported	[PMID 9600981; 9748275]
PaaH is cytoplasmic/soluble, likely homodimeric	Supported (inference)	No signal/TM; family structures [PMID 24792376, 10840044]
PaaH is a membrane transporter or structural/signaling protein	Refuted	Sequence and family are enzymatic, soluble oxidoreductase
The gene symbol is ambiguous / literature belongs to a different gene	Refuted	All identifiers converge on the phenylacetate-pathway HAD

7. Limitations and Future Directions

- **No direct enzymology on Q88HS1 itself:** substrate specificity and kinetic parameters are inferred from pathway context and close homologs, not measured for the KT2440 protein.

Direct assay with (S)-3-hydroxyadipyl-CoA vs. straight-chain and other dicarboxylic 3-hydroxyacyl-CoAs would establish specificity and any promiscuity.

- **No experimental structure of Q88HS1:** the two-domain fold, dimerization, and His-Glu dyad positions are inferred from RePaaH1 and human HAD. A crystal/AlphaFold structure would localize the catalytic residues and confirm cofactor geometry.
- **Nomenclature mapping:** cross-strain *paa* gene-letter differences warrant care; confirming PP_3282 operon context and coexpression with the other *paa* genes in KT2440 would solidify the assignment.
- **Regulation:** the pathway is induced by phenylacetyl-CoA via a PaaX-type repressor in *E. coli*; verifying PaaH's transcriptional control in KT2440 would complete the physiological picture.

Key References

- Olivera et al. 1998, *PNAS* — *P. putida* U phenylacetyl-CoA catabolon; β -oxidation-like system
([P 9600981](#))
- Ferrández et al. 1998, *J Biol Chem* — *E. coli* paa cluster; β -oxidation-like enzymes
([P 9748275](#))
- Nogales et al. 2007, *Microbiology* — terminal step; succinyl-CoA + acetyl-CoA products
([P 17259607](#))
- Luu et al. 2013, *Appl Environ Microbiol* — PAA as funneling pathway / growth substrate
([P 23377939](#))
- Kim et al. 2014, *Biochim Biophys Acta* — PaaH1 (*Ralstonia*) crystal structure; Rossmann + dimerization domains
([P 24792376](#))
- Barycki et al. 2000, *Biochemistry* — L-3-hydroxyacyl-CoA dehydrogenase mechanism, His-Glu dyad
([P 10840044](#))

- Arias et al. 2008

(P 18177365);

Alonso et al. 2003

(P 14597173)

— upstream catabolon routes converging on phenylacetyl-CoA

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